

## ST ANDRAE LAVANTTAL, Austria

WMO: 112290

Lat: 46.764N

Lon: 14.828E

Elev: 1324

StdP: 14.01

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 8.9        | 13.1 | 5.8                           | 7.8 | 11.2 | 9.8 | 9.5 | 16.4 | 12.9                  | 33.2 | 10.9 | 34.3 | 1.5                   | 20   | 0.459 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.8                   | 88.3            | 70.4 | 85.0 | 69.4 | 82.0 | 68.0 | 72.3                | 84.8 | 70.6 | 82.5 | 69.0 | 79.9 | 3.6                  | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 68.0                            | 108.2 | 79.0 | 66.4 | 102.2 | 76.5 | 65.0 | 97.1 | 74.4 | 36.8          | 85.1 | 35.2 | 83.1 | 33.9 | 80.1 | 78.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |       |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|-------|
| 1%                | 2.5% | 5%  | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |       |
|                   |      |     | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |       |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |       |
| 11.6              | 8.9  | 6.7 | DB                         | 3.9 | 93.5               | 6.9 | 2.9                                                | -1.0 | 95.6       | -5.0 | 97.3       | -8.9 | 98.9       | -13.9 | 101.0 |
|                   |      |     | WB                         | 3.5 | 75.2               | 6.8 | 1.6                                                | -1.3 | 76.4       | -5.3 | 77.4       | -9.1 | 78.3       | -14.0 | 79.5  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 49.6   | 28.3  | 33.3  | 41.9  | 50.6  | 58.4  | 65.7  | 68.6  | 67.3  | 58.8  | 49.7  | 40.4  | 30.5  | (6)  |
|                                                                        | DBStd       | 15.29  | 7.04  | 6.92  | 6.77  | 5.94  | 6.02  | 6.24  | 5.65  | 5.27  | 5.75  | 6.60  | 6.80  | 6.30  | (7)  |
|                                                                        | HDD50       | 2455   | 673   | 468   | 256   | 64    | 6     | 0     | 0     | 0     | 2     | 88    | 294   | 604   | (8)  |
|                                                                        | HDD65       | 6004   | 1138  | 888   | 715   | 431   | 218   | 66    | 30    | 37    | 200   | 474   | 738   | 1069  | (9)  |
|                                                                        | CDD50       | 2291   | 0     | 0     | 6     | 82    | 266   | 471   | 577   | 537   | 267   | 79    | 6     | 0     | (10) |
|                                                                        | CDD65       | 365    | 0     | 0     | 0     | 0     | 12    | 87    | 143   | 109   | 14    | 0     | 0     | 0     | (11) |
|                                                                        | CDH74       | 3585   | 0     | 0     | 0     | 16    | 245   | 896   | 1355  | 954   | 117   | 2     | 0     | 0     | (12) |
|                                                                        | CDH80       | 1098   | 0     | 0     | 0     | 0     | 42    | 267   | 482   | 296   | 11    | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 2.6    | 2.2   | 2.5   | 3.0   | 3.3   | 3.2   | 2.8   | 2.8   | 2.4   | 2.3   | 2.4   | 2.3   | 2.1   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 35.5   | 1.0   | 1.2   | 1.5   | 2.3   | 3.6   | 4.5   | 5.0   | 5.0   | 3.8   | 3.2   | 2.6   | 1.7   | (15) |
|                                                                        | PrecMax     | 45.0   | 2.7   | 4.4   | 3.5   | 4.7   | 5.6   | 8.1   | 11.7  | 9.1   | 6.0   | 7.2   | 6.4   | 3.8   | (16) |
|                                                                        | PrecMin     | 26.6   | 0.2   | 0.0   | 0.2   | 0.6   | 1.9   | 2.6   | 1.1   | 1.7   | 1.9   | 0.0   | 0.1   | 0.1   | (17) |
|                                                                        | PrecStd     | 4.4    | 0.6   | 1.1   | 0.9   | 1.0   | 1.3   | 1.4   | 2.2   | 1.8   | 1.1   | 1.8   | 1.5   | 1.1   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 48.4  | 57.2  | 68.5  | 76.3  | 84.8  | 90.1  | 93.3  | 92.1  | 81.5  | 73.0  | 60.6  | 48.0  | (19) |
|                                                                        |             | MCWB   | 42.8  | 46.9  | 52.7  | 59.2  | 66.0  | 70.4  | 72.5  | 70.9  | 67.7  | 61.5  | 53.6  | 44.0  | (20) |
|                                                                        | 2%          | DB     | 42.6  | 50.8  | 63.9  | 72.1  | 80.0  | 86.2  | 89.1  | 86.6  | 77.4  | 67.9  | 55.8  | 43.1  | (21) |
|                                                                        |             | MCWB   | 38.8  | 42.8  | 51.1  | 56.1  | 63.6  | 69.4  | 71.3  | 70.0  | 65.6  | 58.6  | 51.9  | 40.5  | (22) |
|                                                                        | 5%          | DB     | 39.5  | 46.4  | 59.8  | 68.7  | 76.4  | 83.1  | 85.4  | 83.1  | 73.8  | 64.2  | 52.6  | 40.4  | (23) |
|                                                                        |             | MCWB   | 36.7  | 40.3  | 49.1  | 54.9  | 61.9  | 68.1  | 69.9  | 69.4  | 63.5  | 56.8  | 49.9  | 38.8  | (24) |
|                                                                        | 10%         | DB     | 36.8  | 42.7  | 55.3  | 64.6  | 72.4  | 79.5  | 81.9  | 79.4  | 69.9  | 60.2  | 49.9  | 38.3  | (25) |
|                                                                        |             | MCWB   | 34.8  | 38.3  | 46.4  | 53.0  | 59.9  | 66.3  | 68.4  | 67.7  | 61.5  | 54.7  | 48.2  | 37.0  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 43.2  | 47.3  | 53.9  | 60.2  | 67.6  | 73.2  | 74.5  | 74.3  | 68.6  | 62.3  | 55.0  | 45.4  | (27) |
|                                                                        |             | MCDB   | 46.1  | 56.3  | 65.6  | 74.8  | 80.2  | 86.2  | 88.8  | 85.2  | 78.8  | 71.3  | 58.3  | 47.2  | (28) |
|                                                                        | 2%          | WB     | 39.8  | 43.5  | 51.6  | 57.3  | 65.1  | 70.8  | 72.5  | 71.9  | 66.4  | 59.7  | 52.6  | 41.3  | (29) |
|                                                                        |             | MCDB   | 42.2  | 49.8  | 62.7  | 69.1  | 77.5  | 83.5  | 86.3  | 82.8  | 75.3  | 66.4  | 55.3  | 42.4  | (30) |
|                                                                        | 5%          | WB     | 37.1  | 40.9  | 49.5  | 55.5  | 62.9  | 68.9  | 70.7  | 70.2  | 64.5  | 57.6  | 50.3  | 39.1  | (31) |
|                                                                        |             | MCDB   | 38.8  | 45.1  | 58.7  | 66.3  | 74.6  | 80.5  | 83.2  | 81.2  | 72.2  | 63.2  | 52.2  | 40.1  | (32) |
|                                                                        | 10%         | WB     | 35.0  | 38.7  | 47.2  | 53.8  | 60.6  | 66.9  | 69.1  | 68.4  | 62.4  | 55.3  | 48.2  | 37.2  | (33) |
|                                                                        |             | MCDB   | 36.4  | 42.2  | 54.2  | 63.4  | 70.3  | 77.5  | 80.0  | 78.1  | 68.3  | 59.5  | 50.1  | 38.1  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.4  | 16.6  | 22.3  | 22.5  | 22.4  | 22.3  | 22.8  | 21.9  | 19.4  | 18.0  | 11.0  | 9.4   | (35) |
|                                                                        |             | MCDBR  | 14.4  | 22.9  | 31.1  | 31.8  | 29.9  | 28.7  | 28.6  | 28.1  | 24.8  | 23.5  | 15.5  | 12.2  | (36) |
|                                                                        |             | MCWBR  | 10.5  | 15.1  | 18.6  | 16.8  | 14.4  | 13.0  | 12.2  | 12.9  | 13.2  | 14.3  | 11.0  | 9.3   | (37) |
|                                                                        | 5% WB       | MCDBR  | 12.3  | 20.9  | 29.6  | 28.2  | 28.4  | 26.7  | 27.0  | 25.9  | 22.4  | 21.7  | 13.5  | 10.4  | (38) |
|                                                                        |             | MCWBR  | 9.4   | 14.2  | 18.0  | 15.1  | 14.0  | 12.7  | 12.3  | 12.7  | 12.1  | 13.6  | 9.7   | 8.3   | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.299 | 0.328 | 0.371 | 0.420 | 0.424 | 0.431 | 0.432 | 0.429 | 0.399 | 0.372 | 0.337 | 0.301 | (40) |
|                                                                        | taud        |        | 2.423 | 2.352 | 2.291 | 2.196 | 2.234 | 2.258 | 2.259 | 2.280 | 2.337 | 2.387 | 2.439 | 2.474 | (41) |
|                                                                        | Ebn at Noon |        | 252   | 266   | 268   | 265   | 268   | 266   | 264   | 259   | 257   | 246   | 234   | 239   | (42) |
|                                                                        | Edh at Noon |        | 22    | 29    | 36    | 43    | 43    | 42    | 42    | 39    | 34    | 28    | 22    | 19    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 399   | 665   | 1022  | 1335  | 1604  | 1784  | 1785  | 1576  | 1128  | 751   | 414   | 328   | (44) |
|                                                                        | RadStd      |        | 53    | 95    | 128   | 148   | 161   | 124   | 131   | 173   | 147   | 82    | 64    | 49    | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (96 neighbors) | +0.92 | N/A     | N/A    | +1.57   | +0.86 | +0.70 | -164        | -247  | +164  | +101  |

Nomenclature: See separate page